Project Management (PM)

Chapter 2 – Project Initiation & Planning

Lesson Contents

- Project Integration Management
- Project Initiation Fundamentals
- Project Planning

Learning Objectives

After completing this lesson, you will be able to

- Describe project integration management processes
- Discuss the strategic planning process and apply different project selection methods
- Explain the importance of creating a project charter to formally initiate projects
- Develop project management plan,

Project Integration Management

Project integration management

- ♦ Includes processes to ensure that all the elements of a project are properly coordinated
- ♦ Making balance among competing alternatives & objectives to meet stakeholder needs
- ♦The Key to Overall Project Success ➔ Good Project Integration Management ➔stakeholder satisfaction

Who Manages Integration?

Project integration management

- ♦ Project Manager – Integrator for a project that executes processes
- ♦ Team Members - concentrate on completing tasks, activities, & work packages
- ♦ Project Sponsor - Protect project from changes & losing resources

Why do we manage Integration

Project integration management

- Manage change & communication
- reduce project time & cost

- Involves stakeholders early & often
- Make results visible

- Identify problems/solutions early
- Use relevant experience as early as possible

- The Key to Overall Project Success ➔ Good Project Integration Management ➔stakeholder satisfaction

Strategic Planning and Project Selection

Integration Management: The Big Picture

- **Think Organization-Wide**
  - Integration management considers the *entire organizational context*, not just a single project.
- **Successful Leaders –** Align projects with organizational strategy.
- **Strategic Planning Includes:**
  - Setting **long-term objectives**
  - **Forecasting trends**
  - **Identifying new product/service needs**
- **As a tool, use SWOT Analysis (**Strengths, Weaknesses, Opportunities, Threats**)**

Identify Potential Projects

- The first step in project management is deciding what projects to do
- **Project initiation** starts here
- The initiation phase sets the foundation
  - defining *why* a project should exist, *who* is involved, and *what* it aims to achieve.
- In software project terms, this is when the idea is validated for **feasibility**, **business value**, and **strategic fit**.
  - I**dentifying** potential projects, using realistic methods
  - Formalizing their initiation by issuing some sort of project charter.

Identify Potential Projects …

1. Identify strategic goals
2. Perform a business area analysis
3. Defining potential software projects, their scope, benefits, and constraints.

- Finally, choosing which projects to do & assigning resources for working on them

Methods for Selecting Projects

focusing on broad organizational needs

- Projects that address broad organizational needs
- There is a ***need*** for the project
- There are ***funds*** available
- There's a strong ***will*** to make the project succeed

categorizing projects

- Impetus
  - **Problems**
  - Opportunities (chances to improve the organization)
  - **Directives (**new requirements imposed by somebody)
- Time window – How long it will take to do & when it is needed
- Overall priority of the project

Financial analysis

- NPV
- ROI
- Payback analysis

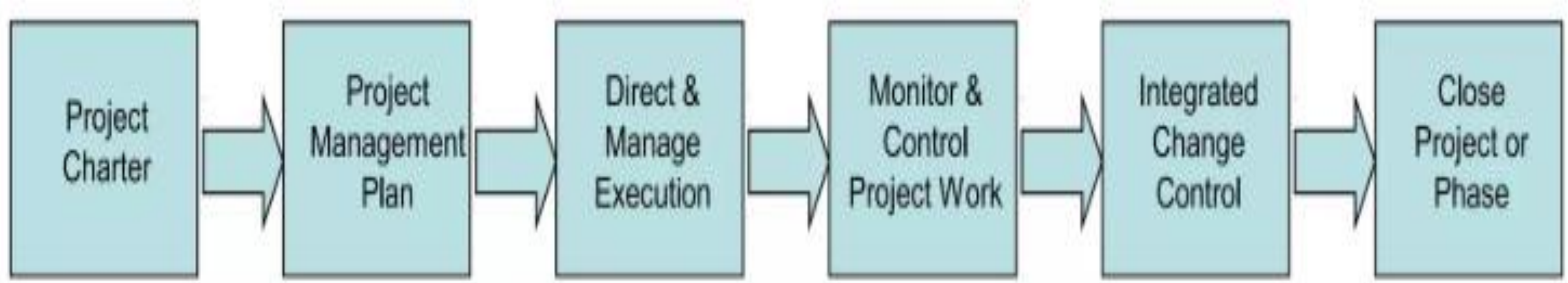
weighted scoring model

- Identify criteria important to the project selection process
- Assign weights & scores to each criterion
- Multiply them to find weighted scores

Read (Kathy Schwalbe, 9th Ed., 2019. , pp. 161 – 169)

How do We Manage Integration?

- Use the seven integration processes
  - Develop project charter
  - Develop project integration management plan
  - Direct & control project work
  - Perform integrated change control
  - **Close project or phase**



- Each process has its own I, T & T, and O

Project Integration Management Processes

(Kathy, 2019. , pp. 153)

Project Integration Management Overview

4.1 Develop Project Charter

.1 Inputs
- .1 Business documents
- .2 Agreements
- .3 Enterprise environmental factors
- .4 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Data gathering
- .3 Interpersonal and team skills
- .4 Meetings

.3 Outputs
- .1 Project charter
- .2 Assumption log

4.2 Develop Project Management Plan

.1 Inputs
- .1 Project charter
- .2 Outputs from other processes
- .3 Enterprise environmental factors
- .4 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Data gathering
- .3 Interpersonal and team skills
- .4 Meetings

.3 Outputs
- .1 Project management plan

4.3 Direct and Manage Project Work

.1 Inputs
- .1 Project management plan
- .2 Project documents
- .3 Approved change requests
- .4 Enterprise environmental factors
- .5 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Project management information system
- .3 Meetings

.3 Outputs
- .1 Deliverables
- .2 Work performance data
- .3 Issue log
- .4 Change requests
- .5 Project management plan updates
- .6 Project documents updates
- .7 Organizational process assets updates

4.4 Manage Project Knowledge

.1 Inputs
- .1 Project management plan
- .2 Project documents
- .3 Deliverables
- .4 Enterprise environmental factors
- .5 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Knowledge management
- .3 Information management
- .4 Interpersonal and team skills

.3 Outputs
- .1 Lessons learned register
- .2 Project management plan updates
- .3 Organizational process assets updates

4.5 Monitor and Control Project Work

.1 Inputs
- .1 Project management plan
- .2 Project documents
- .3 Work performance information
- .4 Agreements
- .5 Enterprise environmental factors
- .6 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Data analysis
- .3 Decision making
- .4 Meetings

.3 Outputs
- .1 Work performance reports
- .2 Change requests
- .3 Project management plan updates
- .4 Project documents updates

4.6 Perform Integrated Change Control

.1 Inputs
- .1 Project management plan
- .2 Project documents
- .3 Work performance reports
- .4 Change requests
- .5 Enterprise environmental factors
- .6 Organizational process assets

.2 Tools & Techniques
- .1 Expert judgment
- .2 Change control tools
- .3 Data analysis
- .4 Decision making
- .5 Meetings

.3 Outputs
- .1 Approved change requests
- .2 Project management plan updates
- .3 Project documents updates

4.7 Close Project or Phase

.1 Inputs
- .1 Project charter
- .2 Project management plan
- .3 Project documents
- .4 Accepted deliverables
- .5 Business documents
- .6 Agreements
- .7 Procurement documentation
- .8 Organizational process assets

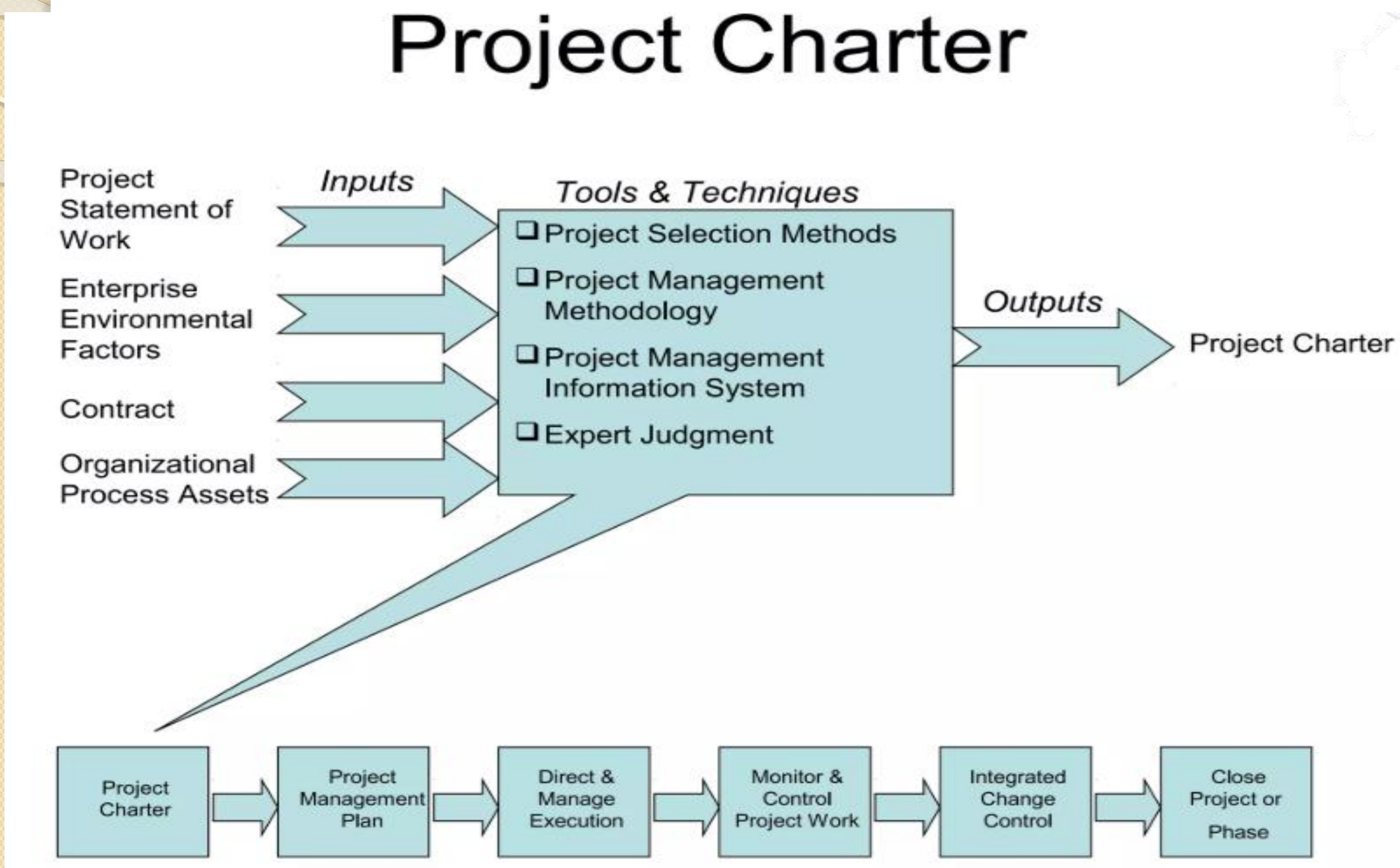
.2 Tools & Techniques
- .1 Expert judgment
- .2 Data analysis
- .3 Meetings

.3 Outputs
- .1 Project documents updates
- .2 Final product, service, or result transition
- .3 Final report
- .4 Organizational process assets updates

Developing a Project Charter

- After top management selects projects, the **organization must be informed** of approved initiative
- **Formal authorization** is required to start — usually through a **Project Charter**
- A **Project Charter**:
  - Officially **recognizes the project's existence**
  - Defines **objectives** and **management direction**
  - **Empowers the Project Manager** to use organizational resources
- **Key stakeholders** must review and **sign** the charter to show agreement on the project's purpose and intent.

Project Charter Input, Tools & Techniques , Output



Inputs for Developing a Project Charter

- **A project statement of work**
  - Describes the products or services to be created by the project team
- **Enterprise environmental factors**
  - Such as relevant government or industry standards, the organization's infrastructure, and marketplace conditions, etc.
- **Organizational process assets**
  - Include formal & informal plans, policies, procedures, guidelines, lessons learned, etc.
- **Agreements**
- contract or agreement (if any) **Organizational process assets**
  - Include formal & informal plans, policies, procedures, guidelines, lessons learned, etc.

Common Components of a Project Charter

- Most common components of project charter include
  - The project's title and date of authorization
  - Project manager's name & contact information
  - A summary schedule & budget of the project
  - A brief description of the project objectives, justification, etc.
  - Project success criteria, including project approval requirements
  - A summary of the planned approach for managing the project – how to manage PMKA & other concerns
  - Roles & responsibility matrixes'
  - A sign-off section & related concerns
  - A comments section in which stakeholders can provide important comments related to the project

Table 2.1: Project Charter for the DNA-Sequencing Instrument Completion Project

Project Title: DNA-Sequencing Instrument Completion Project
Date of Authorization: February 1
Project Start Date: February 1 **Projected Finish Date:** November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated $1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The DNA-sequencing instrument project has been underway for three years. It is a crucial project for our company. This is the first charter for the project, and the objective is to complete the first version of the software for the instrument in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Table 2.2: Project Charter

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the DNA sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (Signatures of all the above stakeholders)

Ahmed Abrams *Nick Carson*
Susan Johnson *Renyong Chi*
Erik Haus *Bill Strom*
Maggie Elliot

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed." —Ahmed Abrams

"The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me." —Renyong Chi

Developing a Project Management Plan

Project Plan (PP)

- Planning is always considered to be the most important Introduction stage in a project life cycle and very much the predecessor to all that takes place in a project through till its completion.
- The planning stage needs to be taken very seriously by the top management, project manager, and the team involved in the project.

Why Does Project Need Plans?

- A plan is almost like a "**MAP**".
- Planning provides many benefits such as

1. Improving Communications with stakeholders
2. Increasing the Transparency of Our Own Work
3. Becoming More Organized and Efficient
   - We will have a clear list of organized activities to do & to follow
4. Focus Lies on the Project Objective
5. Helps to answer "Wh. " questions like
   - What is it trying to resolve? … value will it add to the organization?
   - Why is the project taken up?
   - Who will be involved in the project?

Software Project Plan (SPP)

- PP can be defined as a formal, approved document used to broadly guide both project execution and project control.
- Takes its objective from the project charter and the scope statement.

SPP …

- Questions respond by SPP:
- What?
  - What is it trying to resolve?
  - What value will it add to the organization?
  - What will the output or deliverables of the project be?
  - What are the milestones?
  - What are the scope, budget, and schedule?
- Why?
  - Why is the project taken up?
  - Why was the project undertaken?
- Who?
  - Who will be involved in the project?
  - Who are the team members working on the project? What does each member do while on the project?

Stepwise Project Planning

- There are many approaches of for project planning.
- **Step – Wise – Planning Approach** of Software Project Planning
- This framework illustrates the **basic steps** in project planning and the **various activities involved** in each of the steps throughout the development process.

Stepwise Project Planning …

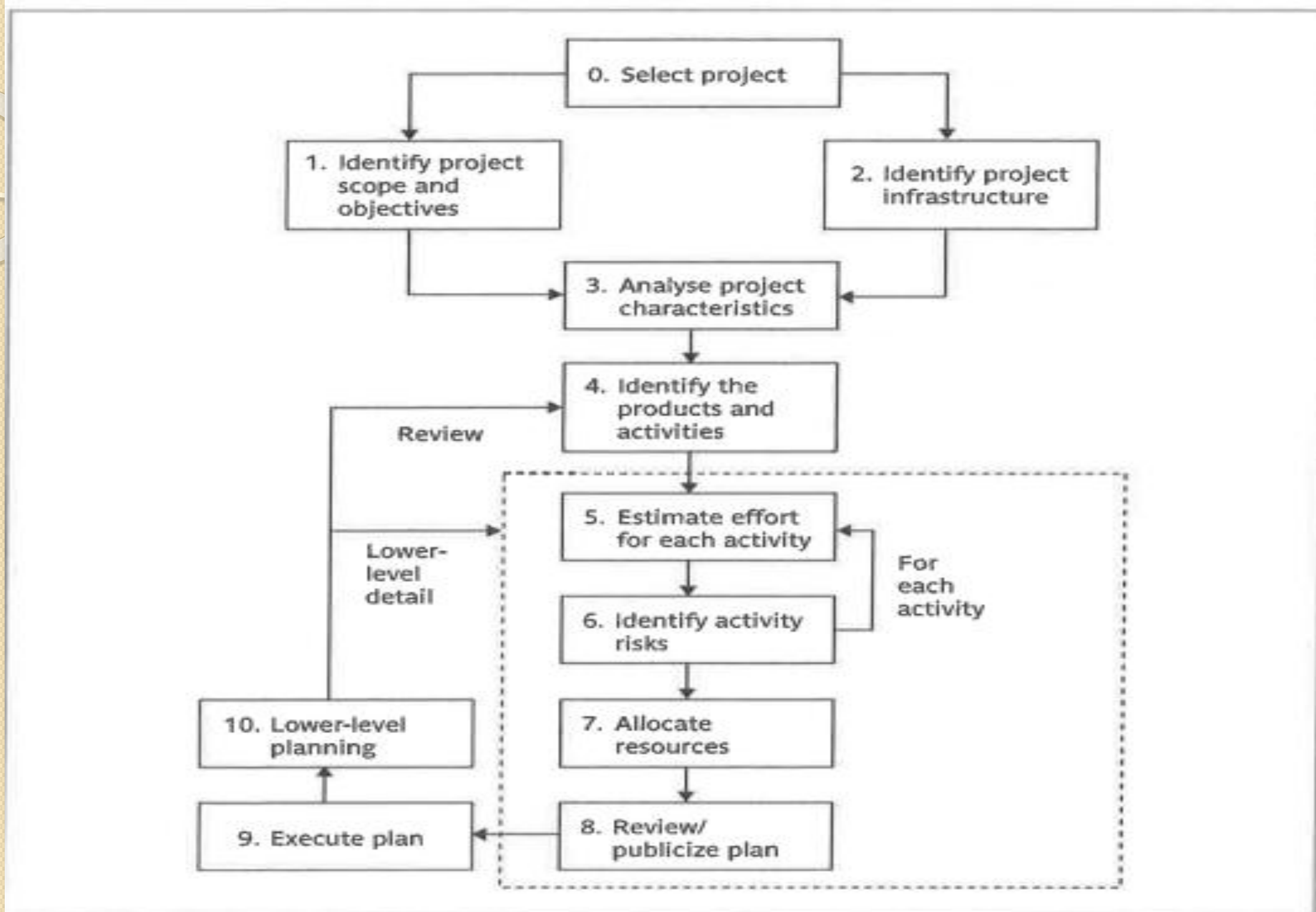


Figure 2.1: Overview of Stepwise Project Planning (Hughes, Bob, 5th ed.)

Developing a Project Management Plan

- A **project management plan** is a central document that coordinates all **project planning components**
- It **integrates subsidiary plans** from other knowledge areas into one comprehensive guide
- A **formally approved document** that defines *how the project work will be executed, monitored, and controlled*
- Provides a **clear roadmap** for managing scope, schedule, cost, quality, risks, and resource
- Ensures **alignment** between project objectives and organizational goals

Develop PMP …

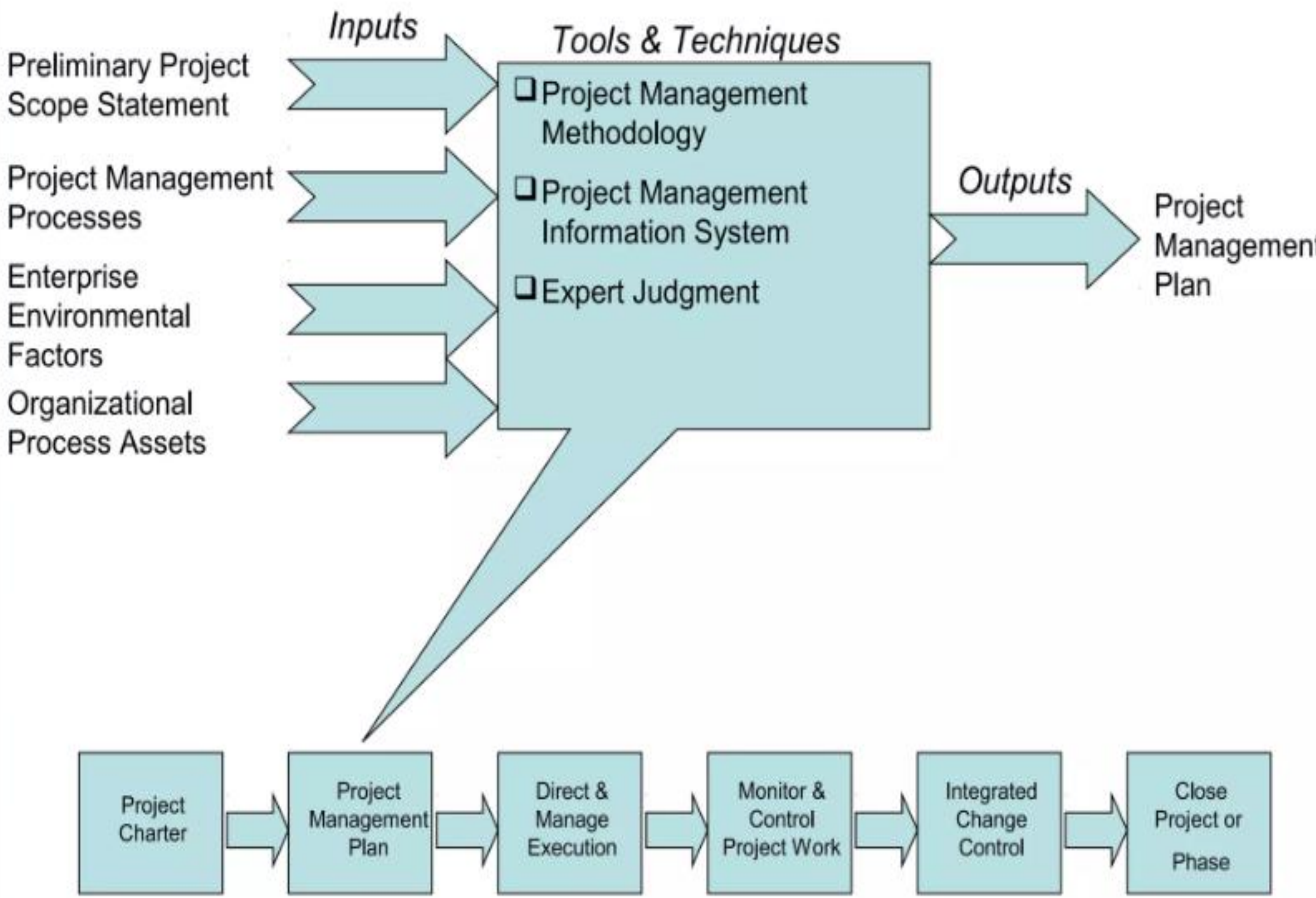
- It includes detailed information on the project's
  - objectives, cost estimates, schedule, resources, staffing, risk management plans, monitoring, control mechanisms, and other related activities.
- By focusing on the "How" aspects of each
  - The plan defines every essential aspect of the project and primarily focuses on the "how"
- For example:
  - How will the project work be executed?
  - How will project management concerns be addressed and managed?, etc.

SPP Vs. SPMP

Key Difference b/n Project Plan & Project Management Plan

Project Plan	Project Management Plan
Deals with high level planning (details the processes required to **perform** the project) ➔low detailed level	Low level view plan (details the processes required to **manage** the project. ➔high detailed level. Defines every possible detail.
Detailed plan about the start of the project through till its end	High detailed plan about **how to start the project & monitor** its every move through till its end
Mainly deals with the **"what"** part of the project.	Mainly deals with the "**how**" part of the project.
A visionary document that plots the project in steps towards the vision or objective	An execution document that plots the activities of the project, the tools & techniques, which will be used, to achieve the vision/objective stated in the project plan
Gives the project manager the vision of what the project steps & stages will be like	Gives the project manager the definitive plan & develops a system to execute the vision of the project plan

Project Management Plan



Common Elements of a Project Management Plan

- Introduction or overview of the project
  - The project name, brief description of the project , sponsor's name, PM manager and key team members, etc.
- Description of how the project is organized
  - Organizational charts, roles & responsibilities
- Management and technical processes used on the project
  - Management objectives, project controls, risk management, methodologies to be used, etc.
- Work to be done, schedule, and budget information
  - Major work packages, key deliverables, summarized & detailed schedule & cost information, etc.

Table 2.3: Sample Contents for a Project Management Plan (PMP)

MAJOR SECTION HEADINGS	SECTION TOPICS
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and Responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition; and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure Plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews & audits; problem resolution plan; subcontractor management plan; process improvement plan

Directing and Managing Project Work

- Involves leading and performing the **tasks defined in the Project Management Plan** to achieve project objectives.
- It is where the **planned work is executed** and deliverables are produced.
  - In simpler terms: It's the phase where the team *actually builds the product* according to the plan.
  - **It transforms plan to action**
- **Purpose**
  - Implement the activities outlined in the project management plan
  - Produce deliverables and project outcome
  - Ensure project objectives are being me
  - Provide input for monitoring and controlling processes

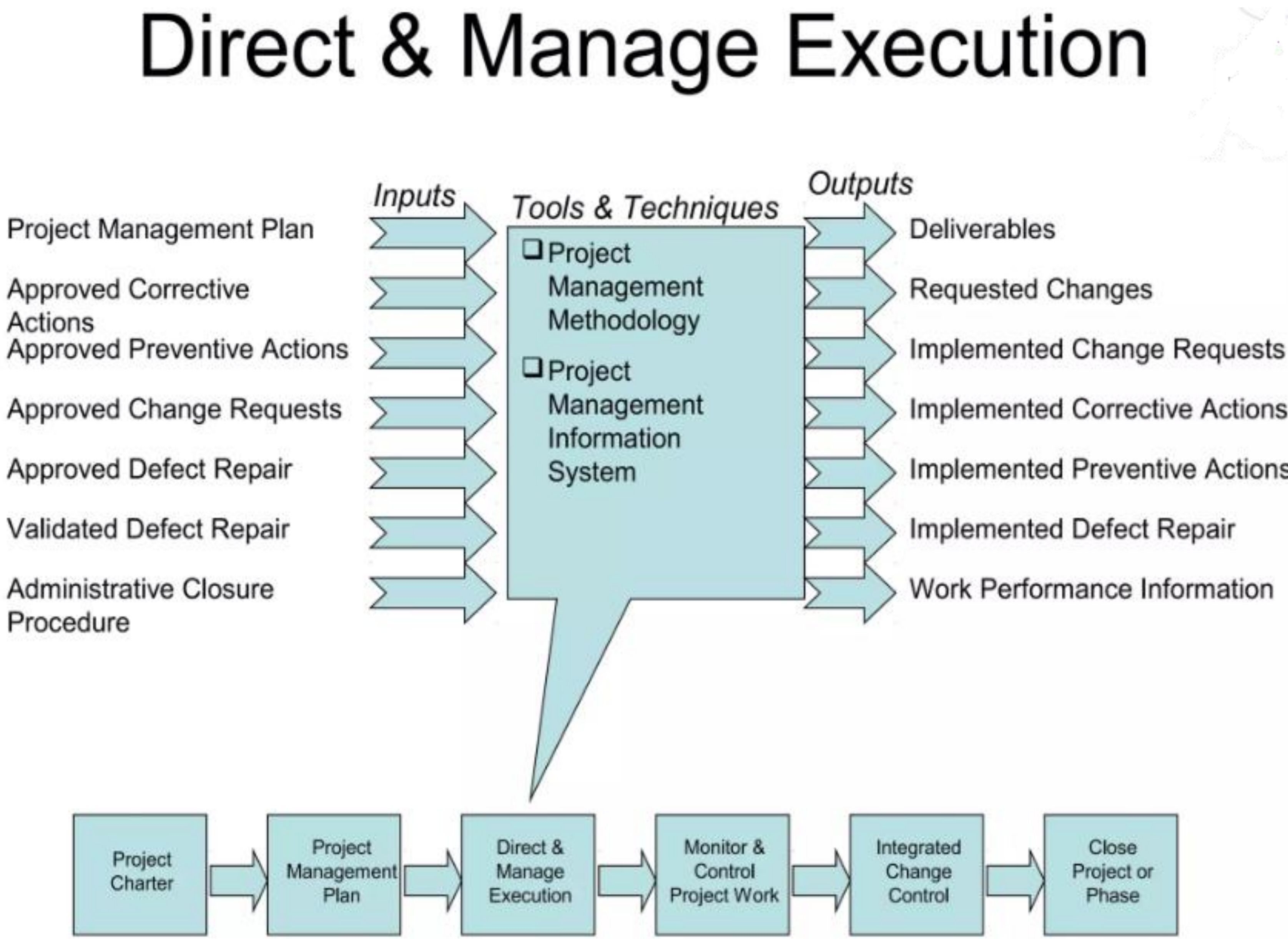
Key Activities

- Executing Work Package
- Coordinating People & Resources (e.g. assign tasks, check the availability of proper tools & environment, etc.)
- Implementing Approved Changes
- Producing Deliverables (e.g. software modules, documentation, reports, etc.)
- Providing Work Performance Data
- Managing Project Knowledge (e.g. lessons learned, decisions, best practices, etc.)

Key considerations

- Ensure all activities align with project objective
- Team coordination is critical to avoid duplication or gaps
- Use PMIS or project tools (Jira, MS Project, etc.) for tracking
- Document lessons learned for future projects

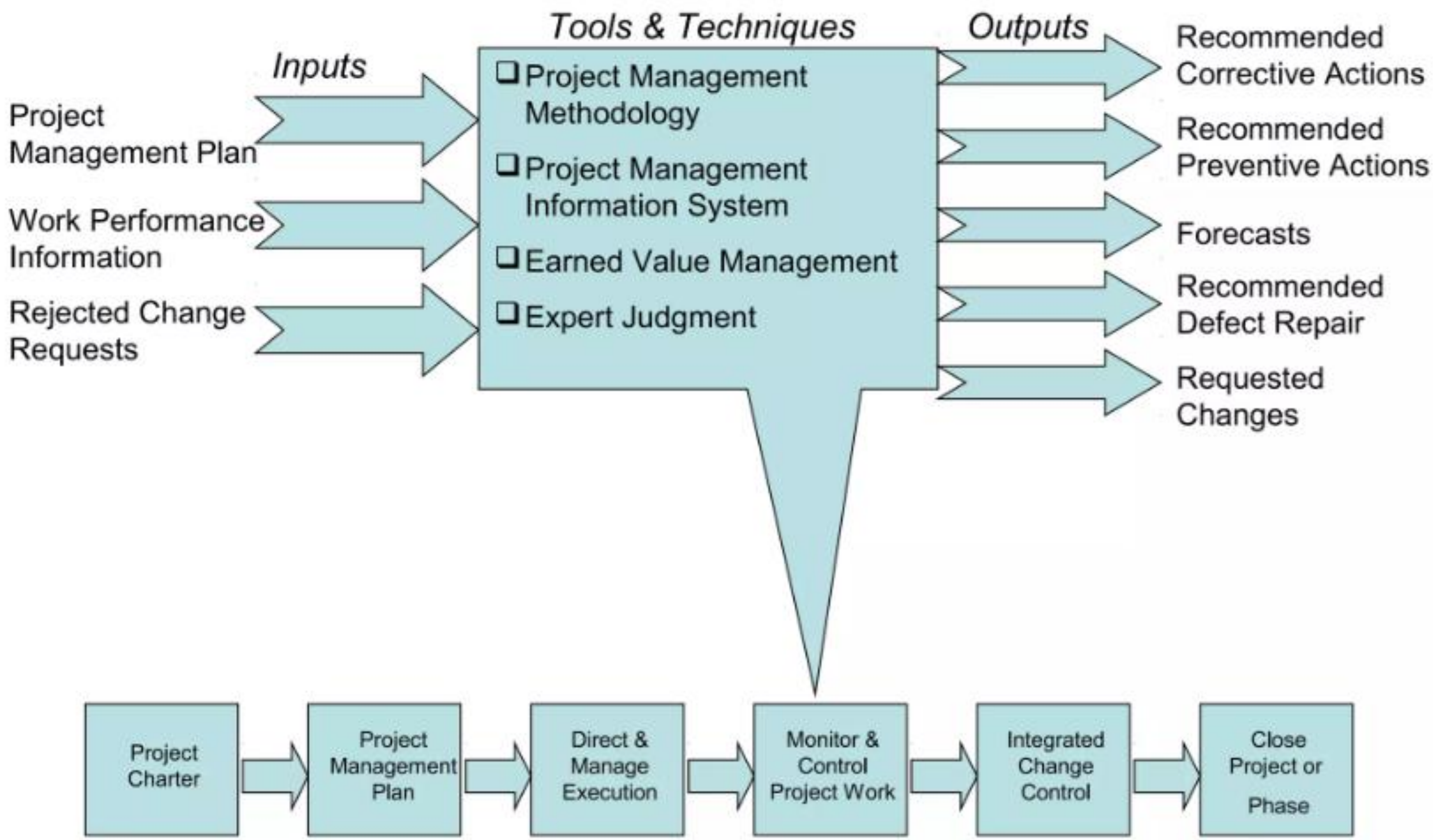
Direct & Manage Execution – I, T & T, O



Monitoring and Controlling Project Work

- Changes are inevitable on most projects, so it's important to develop and follow a process to monitor and control changes
- Involves collecting, measuring, and disseminating performance information
- The project management plan provides the **baseline** for identifying and controlling project changes.
  - **Baseline** = approved PMP + approved changes

Monitor & Control Project Work



Performing Integrated Change Control

- Integrated change control involves **identifying, evaluating, & managing changes throughout the project life cycle.**

Change Control System

- A **change control system** is a formal, documented process that describes when & how official project documents and work may be **changed.**
- Describes who is authorized to make changes and how to make them

Change Control Board (CCB)

- A **change control board** is a formal group of people responsible for approving or rejecting changes on a project
- Includes stakeholders from the entire organization

CCBs provide guidelines for

- preparing change requests,
- evaluate change requests, and
- manage the implementation of approved changes

Performing Integrated Change …

- **Making Timely Changes**
  - Delegate changes to the lowest level possible, but keep everyone informed of changes

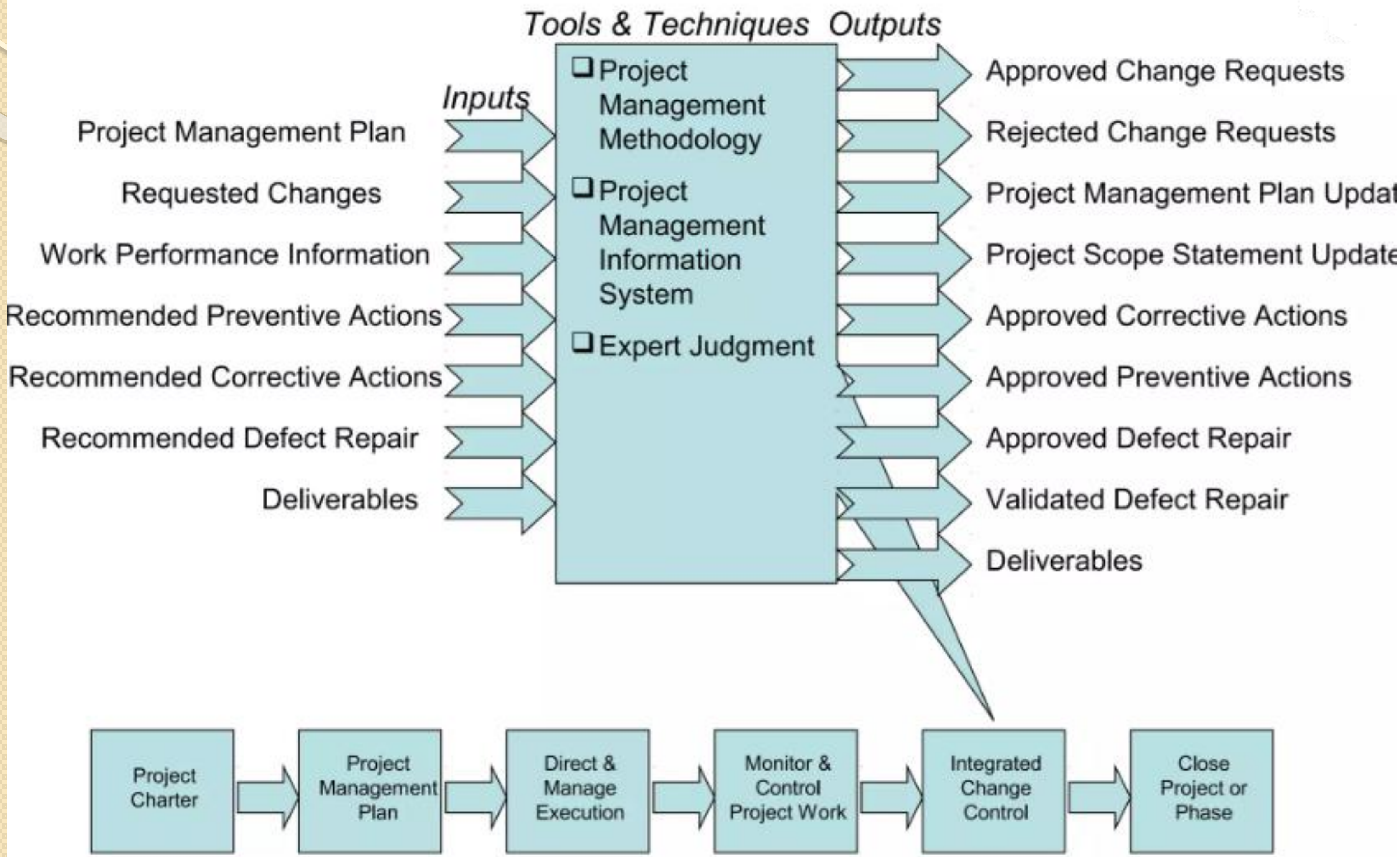
Configuration management

- ensures that the descriptions of the project's products are correct and complete.
- Involves
  - identifying and controlling the functional and physical design characteristics of products and their support documentation.
- Configuration management specialists identify and document
  - configuration requirements, control changes, record and report changes, & audit the products to verify conformance to requirements.

Table 2.4: Suggestions for Performing Integrated Change Control

View project management as a process of constant communication and negotiation.
Plan for change.
Establish a formal change control system, including a **change control board (CCB)**.
Use effective configuration management.
Define procedures for making timely decisions on smaller changes.
Use written and oral performance reports to help identify and manage change.
Use project management and other software to help manage and communicate changes.
Focus on leading the project team and meeting overall project goals and expectations.

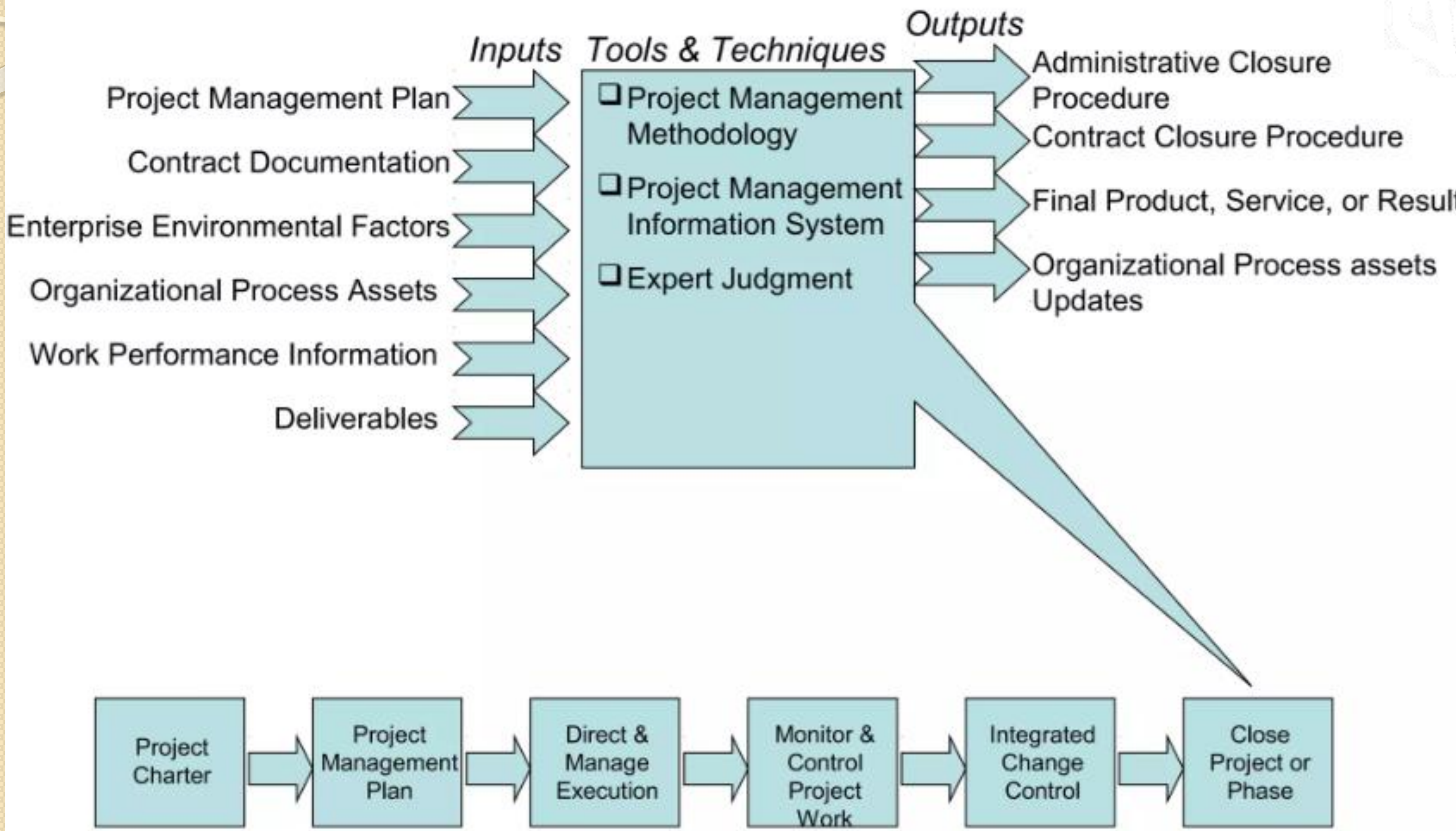
Integrated Change Control



Closing Projects or Phases

- To close a project or phase, you must finalize all activities & transfer the completed or cancelled work to the appropriate people
- Main outputs include
  - Final product, service, or result transition
  - Organizational process asset updates
    - A list of project documentation, project closure documents, and historical information produced by the project in a useful format
  - A final project report and presentation are also commonly used during project closing
- **Notice:**
  - **You can use Software to Assist in Project Integration Management**

Close Project or Phase



Summary of Integration Management

What Goes in a Project Plan?

- Project charter
- Project Management Approach
- Scope Statement
- Work Breakdown Structure (WBS)
- Responsibility Chart
- Network Diagram with Major Milestones
- Budget
- Schedule
- Resources
- Change Control System
- Performance Measurement Guidelines
- Management Plans (Scope, Schedule, cost, quality, staffing, communications, risk response, procurement)

Summary of Integration Management

PM Responsibility for Change

- Influence factors that affect change
- Ensure change is beneficial
- Determine if a change has occurred
- Determine if a change is needed
- Look for alternatives to change
- Minimize negative impact from change
- Notify Stakeholders impacted by change
- Managing those changes that do occur according to project plan

Project Work

- Identifying real world problems that you are interested in & selecting the potential one
- Initiate the project
  - defining why the selected project is important, who is involved, and what it aims to achieve.
- Prepare draft project management plan that will be updated throughout the processes.

Lesson Summary

- Project integration management the heart that ties together all the other areas of project management and process involved.
- When you think to do a project, see the big picture
  - To be successful, our project/s should align and support the strategical plan of a given organization.
- Have good PMP & performing main processes of integration mgt. helps us to handle all aspects of a project, including change mgt.
- Clearly identifying, defining, validating, and controlling using suitable tools and techniques is a tool to avoid scope creeping, project failure, dalliance, and other related problems.
- Try to make your project's scope statement detailed and up to date periodically.

Thank You